--- name: water-service-pareto description: Create or refresh a verified Pareto Chart of Northstar water-service overdue backlog by a requested grouping column. ---

Water-service Pareto Chart

Required inputs

Require the actual source table, report date, selected scope, grouping column and count metric. Ask for missing inputs. In this course use tblWorkOrders, Control!B2 (2026-09-30), all sites and overdue-backlog count grouped by Job_Category or Site. Do not assume access to a web chat or another workbook.

Source and workbook rules

Read the visible .Rules worksheet in column A. Preserve Work_Orders, Technicians, Control, Prompt_Log, .Rules, source IDs, statuses, owners and unrelated sheets. Resolve conflicting requirements explicitly; do not promise a universal precedence order or treat instructions as a security boundary.

Method

Count rows where Status is neither Closed nor Cancelled and Due_Date is strictly before the report date. Group by the requested column. Sort count descending, then group label alphabetically for ties. Build a formula-backed Count, Cumulative_Count, Share and Cumulative_Percent summary. The last cumulative percentage is 100% when the total is positive; for zero total show N/A and explain there is no Pareto distribution.

Output

Create only Pareto_<grouping column>, for example Pareto_Job_Category or Pareto_Site. Ask before replacing an existing matching output. Rebuild that output after review without duplicating summaries or charts. Preserve all other sheets and source records. Create an editable Excel combo chart: descending count columns on the left axis and cumulative percentage line on the right axis fixed at 0%-100%. Label both axes, grouping, metric, date and scope. If the combo UI is unavailable, supply the correct summary and explain the manual combo-chart route; do not claim the chart exists.

Checks

Reconcile the summary count to the eligible source rows. Check stable tie ordering, monotonic cumulative percentages and final 100%. Identify the first category set that reaches at least 80%; do not force a literal 80/20 split. Counts show contribution, not root cause, safety priority or approved actions. A rerun must preserve source records and unrelated outputs.